

TOPICAL REVIEW

Means of Employing Radiation Nulls in Filtennas: A Review

A. VEERABHADRA RAO¹, (Graduate Student Member, IEEE),

AND GOPI RAM¹, (Senior Member, IEEE)

Department of Electronics and Communication Engineering, National Institute of Technology Warangal, Hanamkonda 506004, India

Corresponding author: Gopi Ram (gopi.ram@nitw.ac.in)

ABSTRACT In this paper, methods for improving frequency selectivity in filter antennas by employing controllable radiation nulls (RN) are reviewed. Radiation nulls in the Gain graph help in improving the rejection of out-of-band frequencies. The majority of the designs reviewed in this paper use substrate-integrated waveguide (SIW) technology. Some of the methods of generating RNs include: loading filtenna with slots or shorting pins, using proper design of multi-signal path between source and load, employing electric and magnetic mixed-coupling, embedding filtering structures, parasitic elements, employing filtering action in the feed network. Certain designs achieve more than two radiation nulls which give even better selectivity.

INDEX TERMS SIW, filtenna, radiation null, out-of-band rejection, selectivity.

I. INTRODUCTION

The number of applications of substrate-integrated waveguides [1] has increased over the years. The technology can not only be used in filter design [2], [3], but also in the design of filtennas [4], [5], [6], [7], duplexers [8], [9], power dividers [10], multiplexers, phase shifters [11], etc.

Filter design at mm-wave and microwave frequencies usually includes two or more cavities that are inter-coupled. Instead of increasing the order of the filter to increase the selectivity of frequency response, one could introduce direct coupling between cavities, and cross-couplings between non-adjacent cavities in such a way that transmission zeros are generated on either side of pass band. But when the number of cavities increases, volume increases. So, increasing the cavities number is not suggested. Means of achieving multi-cavity-like operation using a lesser number of cavities have come.

Integration of various components results in multifunction operations like filtenna, etc. A filtenna, which is the integration of filter and antenna, using substrate-integrated waveguide technology helps in reducing losses as well as

achieving more compact designs, especially at mm-wave frequencies.

A simple mechanism of converting a SIW filter to a filtenna is by introducing a radiation slot in one of the cavities of the filter. The size of the filtenna can be further reduced, to half, by using half-mode SIW operation. Other ways of converting a filter to filtenna are by embedding filtering structures, parasitic elements, etc into antenna elements, without the need for an extra filtering circuit. Another construction principle of filtenna is to use an antenna element as the load to the filter circuit.

Some of the other performance indicators of filtenna include compactness, frequency selectivity, wide-band operation, multi-band operation, and reduction of mutual coupling between antenna elements. In this paper, some of the filtenna designs in the existing literature that focus on improving frequency selectivity by introducing radiation nulls/ transmission zeros are discussed.

Compactness could be achieved by using half mode SIW [12], quarter mode SIW [13], substrate-integrated folded waveguide [14], [15], [16], based on negative order resonance [17], utilizing split ring resonators (SRR) and complementary SRR [3], [18], [19], using combine filter [20], using dual-mode cavity [21], [22], increasing the effective capacitance by use of a patch shorted to one of the cavity

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walls of a dual layer cavity resonator [23], by suspended strip line [24], etc.

Radiation nulls represent transmission zeros in a filter or negligible gain points in the Gain graph. They could be carefully located on lower and upper band edges to obtain good selectivity. They help in improving out-of-band suppression. An even more advantageous feature is having independently tunable radiation nulls. There exist different means of introducing RNs in a design. Multiple radiation nulls could be achieved by mixing one or more of these techniques. Some of these techniques are briefly discussed one by one here.

A. EMPLOYING SLOTS/SLITS OF VARIOUS SHAPES

By employing slots/slits of various shapes on the filtenna structure one could get RNs.

The shape of the slots could be straight, open loop/split ring resonator (SRR), Complementary SRR, or U-shaped. In [4], the design uses a U-shaped slot to achieve one RN. In [25], two RNs are achieved using two pairs of slots. Positioning of the slots is chosen such that at the frequency of RN, surface currents on either side of the slots flow in opposite directions causing little or no radiation [25, p. 826]. In [26], a complementary split ring resonator is used on the bottom side of the cavity, which helps in the generation of a RN. Two RNs are achieved using a single slot in [27] with the help of shorting pins. In addition to getting RNs on both sides of the pass band, two more nulls are present to improve frequency selectivity [28]. Using dual slots in each of the dual substrate integrated cavities with unequal spacing in each case, two RNs are achieved [29]. In [30], two RNs are achieved using two pairs of slots, one longer and one shorter. In [31] and [32], one RN is achieved by using straight slots. In [33], a U-shaped slot is used whereas in [34], a split ring slot is used to generate RN. The antenna designs present in [28], [29], [30], and [34] are discussed in sections II-A, II-B, II-C, and II-D respectively.

B. USING SHORTING PINS/VIAS/METALLIC POSTS

Shorting pins/vias/metallic posts could be placed at convenient positions in the cavities such that the surface current and fields could be altered and RN is achieved in the gain graph. The distance of shorting pins from the center axis could be changed to control the position of the RN.

Three shorting pins are loaded in the designs of [4] and [27]. They assist the slots in providing better-filtering action by giving better suppression outside the pass band. Metalized posts in [31] help in increasing the order of the filter, which is an important characteristic. In [35], shorting pins used in single-layer dual-band HMSIW filtenna present not only transmission zeros but also introduce a new resonance. One RN is achieved using two pairs of shorting posts [36]. Selectivity in the lower stopband has been enhanced using nine metallic pins [37]. Details of the design used in [31] are seen in section II-E.

C. USING MULTIPLE COUPLING PATHS

Multiple coupling paths could be developed between the source/load and resonator/antenna by adding extra elements to the antenna structure. A main path and one of the additional paths have a path phase difference of 180 degrees means the far field is canceled, yielding a radiation null. There exists multi-path coupling and the associated coupling coefficients in the coupling matrix are properly chosen to have opposite phases.

Multi-path coupling is achieved using two open loop slots to give rise to more number of RNs [38]. In [39], the use of a U-shaped strip, square rings, and split ring slots gives multiple RNs. In [40] and [41], a coupling diagram is used for setting the required amount of coupling between inter-cavities and ports to improve the frequency selectivity. Two transmission zeros are developed using a dual R-probe feed in the filter antenna [42]. Metal-insulator-metal capacitors are used in [43] and because of resonating mode and nearby modes sharing multiple coupling paths, three RNs are formed. In [44], the filtenna includes a mushroom structure and a slot to control RNs. In [45] two RNs form due to the introduction of a new coupling line by use of an H-shaped aperture. The designs used in [43] and [44] are reviewed in sections II-F and II-G respectively.

D. WITH ELECTRIC AND MAGNETIC MIXED-COUPLING

A transmission zero is formed by employing structures in filtenna that provide both Electric and Magnetic couplings at the same time [37, p. 838].

Multiple resonant modes within the resonators in [37] are mixed coupled to generate RNs, and hence high selectivity. In [46], fundamental mode resonance and EM mixed-coupling generate lower and upper band edge RNs respectively. In [47], by changing the slot position that controls the coupling between the radiating element and resonator, a controllable RN is made possible. An open loop filter with mixed-coupling also gives RNs [48]. The designs used in [46] and [47] are reviewed in sections II-H and II-I respectively.

E. USING PARASITIC LOOPS, STRIPS, OR STACKED PATCHES

The use of stacked patches to achieve a RN is seen in [4], and [49].

As mentioned in [50, p. 1833], “When a parasitic strip is introduced and placed at the maximum radiation direction of the rectangular loop, the radiation resistance of the antenna will drop sharply at high frequencies. The reflection at the input port will be increased sharply and upper band-edge selectivity is obtained”.

A parasitic strip and loop are used in [32] and [50] respectively for generating RN. In [36] and [51], parasitic patches are used. In [52], a U-shaped slot, a parasitic patch, and a folded T-shaped strip are used to generate four RNs.

Details of the designs used in [32] and [52] are reviewed in sections II-J and II-K respectively.

F. AT FEED NETWORK ITSELF

If frequency selectivity circuitry is used at the feeding network itself it would become RN for the whole filtenna. Techniques like step impedance feeding, a U strip connected to probe feed, using a Stepped impedance resonator (SIR) stub, or Uniform impedance resonator (UIR) on feed could help. SIR stub on feed presents band stop property.

In [25] stepped impedance on feed is used, and a U strip connected to probe feed is used in [34] and [39]. Open circuit stubs are used on feed in [49]. In [51] SIR stub on feed is used. In [53], a stub-loaded resonator is used in the feed network. The design in [54] uses UIR at feed where transmission zero is analyzed by using its equivalent circuit. A patch antenna using combined self-decoupling and filtering properties achieves filtering selectivity [55, p. 1511]. The design in [54] is reviewed in section II-L.

Then we have the other performance indicators: wide-band operation, multi-band operation, and related to mutual coupling.

As seen in [56, p. 8126], coupling between two *L*-shaped probes, that are differentially fed by slot, and the radiating element is chosen for wideband operation. In [57], the cavity-backed SIW slot antenna offers multi-resonance and achieves around 20% bandwidth. Vertical stacking of cavities is employed in [56] and [58]. In [59], dual-mode operation plus the loading effect of shorting pin gives enhanced bandwidth operation. In [60], coupling between modes of operation of SIW resonator structure is properly controlled to achieve increased bandwidth.

In [61] and [62], the use of an X-shaped slot structure gives a dual frequency response. In [63, p.1081], a dual-mode cavity coupled with another slotted cavity gives dual-band operation with orthogonal polarization. In [64], the two operating modes of a square SIW resonator result in dual-band operation by changing the cavity shape. Dual-band SIW with improved radiation characteristics is achieved in [65] and [66]. The coupling mechanism can also be altered to change a single-band response to a dual-band response or can increase bandwidth by adding hybrid modes close to the original frequency band.

Conductor, dielectric, and radiation losses are addressed in [67], and an increase in antenna radiation efficiency is achieved by using a substrate-integrated slab waveguide [68]. In [69], good antenna efficiency is achieved by reducing interference between stacked filters. In [70], the suppression of mutual coupling between slot antenna arrays is discussed.

II. TECHNICAL REVIEW OF SOME DESIGNS

Radiation nulls are observed in the Gain versus frequency graph and they help in improving frequency selectivity. The presence of multiple RNs rejects out-of-band frequencies effectively. They are nothing but transmission zeros in the

S_{21} graph. Different designs have come that achieve good skirt selectivity.

A. BY EMPLOYING MULTIPLE SLOTS ON THE FILTENNA STRUCTURE

The frequency responses of four antenna structures discussed in [28] are given below. Firstly the basic cavity-backed slot antenna is considered. The design is shown in Fig. 1(a). There are no radiation nulls as can be seen in its S_{11} graph shown in Fig. 1(b). The design exhibits non-filtering radiation performance and narrow bandwidth.

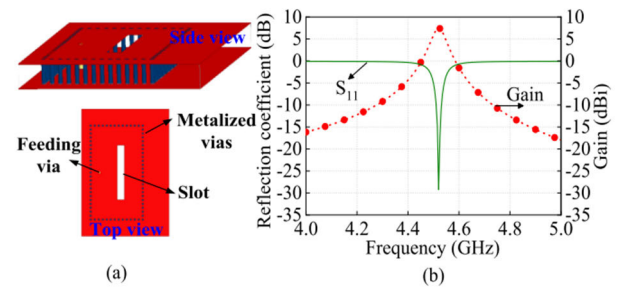


FIGURE 1. First antenna configuration and its Gain plot n [28].

In the modified design of Fig. 2(a), a radiation null has been produced on one side (left) of the pass band as shown in Fig. 2(b). In the next one, see Fig. 3(a), RNs are introduced on both band edges but the suppression levels are not satisfactory as seen in Fig. 3(b). In Fig. 4, different views of the final configuration are shown. The amount of suppression outside the passband is more here because of the introduction of two more RNs. The corresponding Gain plot is shown in Fig. 5.

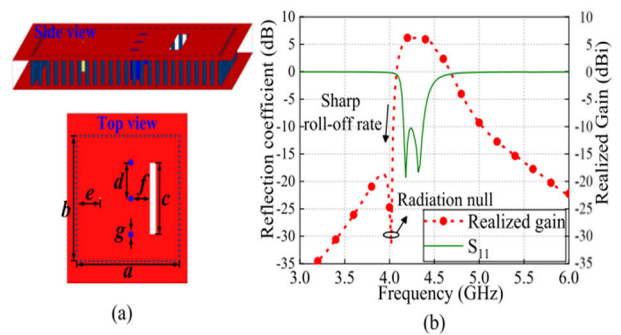


FIGURE 2. Second antenna configuration and its Gain plot [28].

Multiple slots that are used help in controlling the RNs. The slots' positioning, dimensions, and distance between slots are optimized to get tunable RNs.

Fig. 6(a) shows the surface current plot at resonant frequency. Currents on both sides of the slots are almost identical but have opposite directions, so radiation results. Figure 6(b) corresponds to a plot of surface current at one of the RN frequencies. The currents are irregular and weak, currents on both sides are different and so practically speaking, a RN occurs.

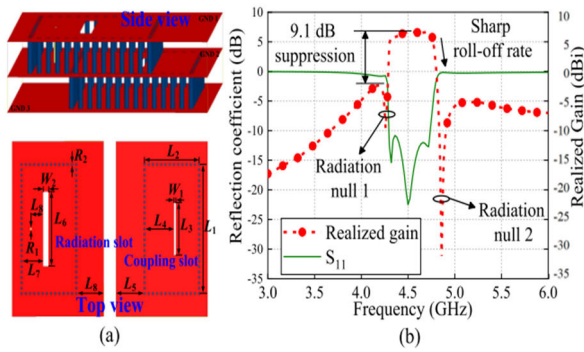


FIGURE 3. Third antenna configuration and its Gain plot [28].

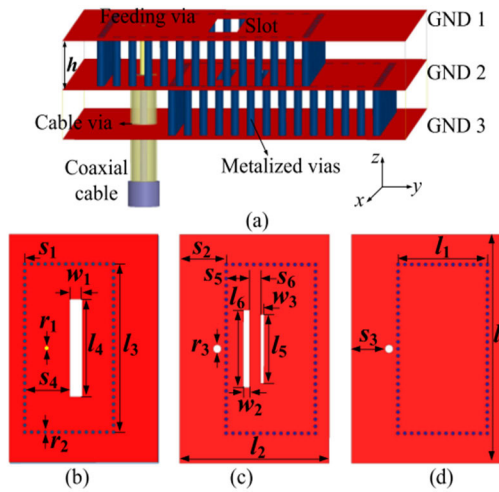


FIGURE 4. (a) 3-D view of the fourth antenna (b) Ground 1 layer with one slot (c) Ground 2 layer with two slots (d) Ground 3 layer [28].

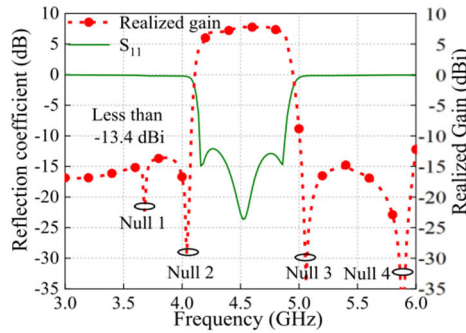


FIGURE 5. S_{11} and Gain plots for the fourth filtenna [28].

B. USING DUAL-SLOTTED CAVITIES

The use of double slotted cavity and the impact of spacing between the slots on the frequency of radiation null as seen in [29] has been reviewed. The design is shown in Fig. 7(a).

Without the dual slots normal TE_{210} mode exists. With the dual slots, there exist two modes: quasi-cavity and quasi-dual slot modes. From [29, p. 1312-1313], for a spacing L_d between the slots that is more than $L/2$, the RN is higher than the two resonant frequencies. If it is less than half the cavity length, $L/2$, it can be seen that the RN frequency lies

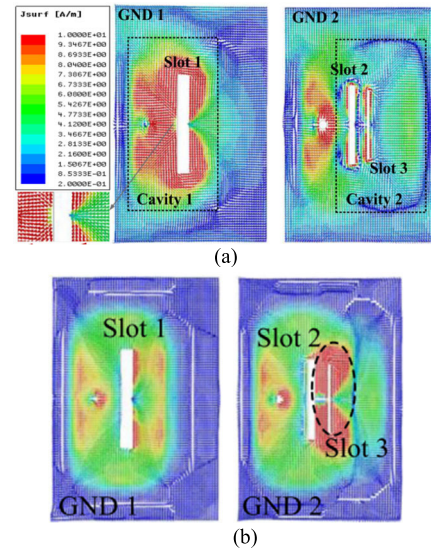


FIGURE 6. Plots of current on both sides of the slot at (a) the resonant frequency and (b) one null frequency [28].

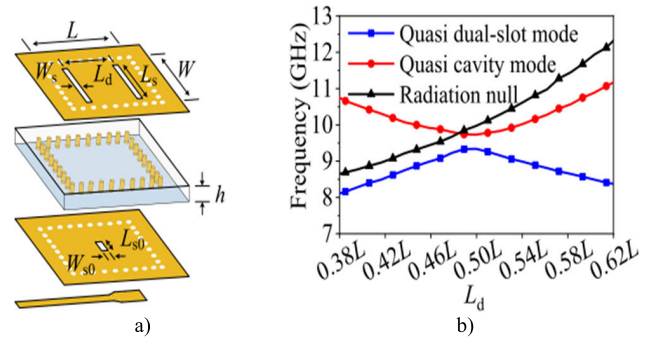


FIGURE 7. (a) Single layer Filtenna (b) plot showing the variation of resonant frequency for different gaps between the slots [29].

in between. The corresponding graph is shown in Fig. 7(b). This logic has been extended to achieve RNs on both sides of the pass band. Two dual-slotted cavities are used, one above the other. The gap between the slots has been set to be less than $L/2$, on one cavity and greater than $L/2$, on the other (shown in Fig. 8), thereby achieving RNs on either side of the passband.

RN is achieved at the frequency where Electric field intensity on both sides of the slot has opposite directions (shown in Fig. 9) thereby yielding no radiation from the slots. Null occurs even though the field there is strong.

As observed by the authors in [29, p. 1313], “When the two-dual slotted Substrate integrated cavities are stacked with $L_{d1} < L/2$ and $L_{d2} > L/2$, the upper and lower resonators produce the upper edge and lower edge radiation nulls, respectively”. Plots of S_{11} and Gain are shown in Fig. 10. Out-of-band rejection of 19 dB is noticed because of the nulls.

C. USING TWO PAIRS OF SLOTS ON THE PATCH

In [30] two pairs of slots are used on the patch (shown in Fig. 11) symmetric with respect to feed location. These slots

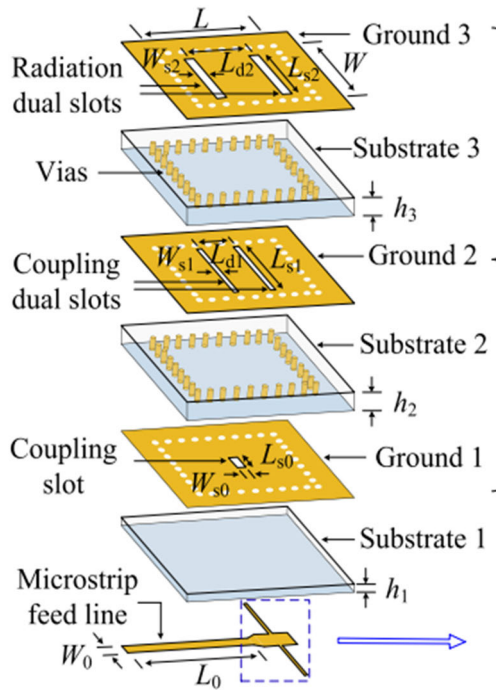


FIGURE 8. Exploded view of the dual layer design [29].

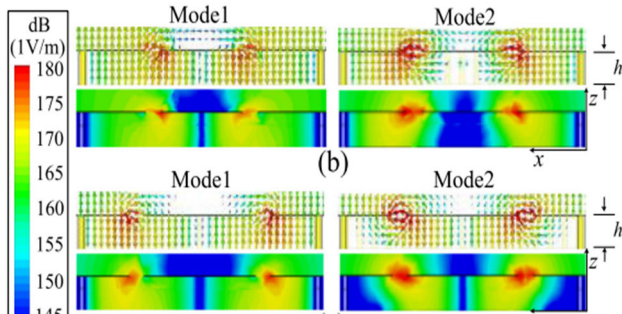


FIGURE 9. The plot of E field around the slots at RN [29].

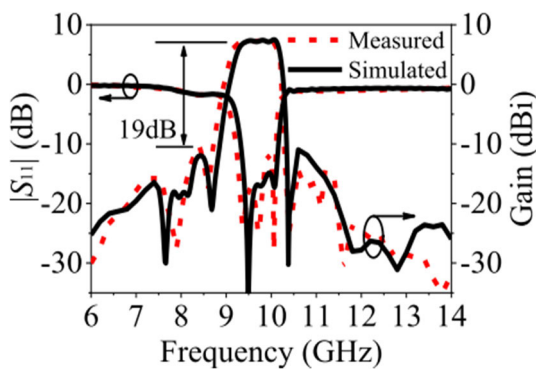


FIGURE 10. S parameter and Gain plots of the design [29].

do not cut currents and so do not disturb performance at the resonant frequency. From [30], one pair (longer slots) gives radiation null towards the lower side of the passband whereas the other pair (shorter ones) gives towards the higher side.

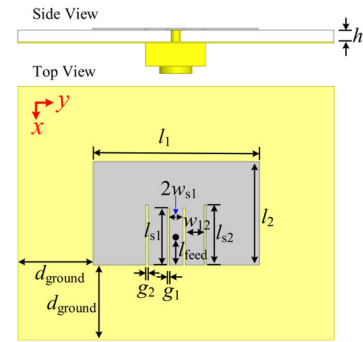


FIGURE 11. Top view of the filtenna with two pairs of slots [30].

Here the efficiency null corresponds to Gain null. Also, the null frequencies are independently tunable by changing respective slot lengths slightly. Confirmation of radiation nulls is done with respect to antenna efficiency, in addition to the antenna gain graph. As shown in the plot of Fig. 12, the presence of RNs is confirmed using both antenna efficiency and Gain curves.

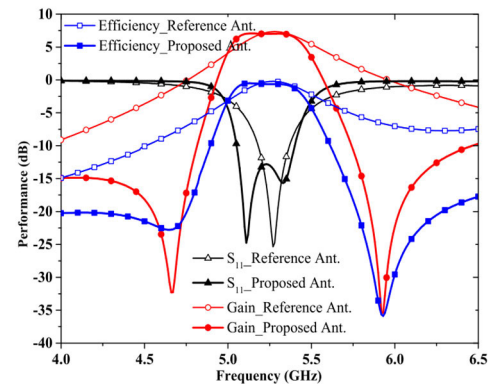


FIGURE 12. S_{11} , Gain, and efficiency curves [30].

D. USING A SPLIT RING-SHAPED SLOT AND U-SHAPED STRIP

A split ring-shaped slot has been used on the patch [34] to achieve RN on the lower side of the passband. The design is shown in Fig. 13(a). From [34, p. 6247], the length of the slot is half of the wavelength at the frequency corresponding to RN. A U-shaped strip connected with feed introduces a RN at the upper band edge. The strip is also chosen to be identically long. In addition to providing RN, the split ring-shaped slot, just like U-shaped slot, generates additional resonance. And so improves impedance bandwidth.

Electric current distribution on the patch, shown in Fig. 13(b), confirms the formation of RN.

The U-shaped strip represents a parallel combination of two quarter-wave open circuit transmission lines which works as a band stop filter [34, p. 6247]. It exhibits a radiation null.

The associated plots of S_{11} and realized Gain are shown in Fig. 14 for different values of the dimension ρ_3 of the split ring-shaped slot. The locations of RNs are seen to be varying.

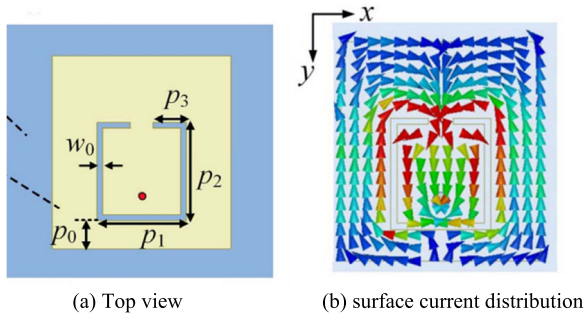


FIGURE 13. SIW cavity with split ring-shaped slot and the surface current plot [34].

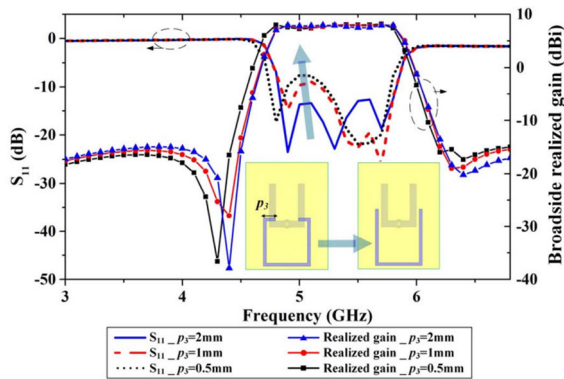


FIGURE 14. S_{11} and Gain plots for different values of the parameter, ρ_3 [34].

E. USING SHORTING PINS

We shall first consider the effect of a single shorting pin on the frequency response of a basic patch antenna. Without the use of a shorting pin, the current density plot on a patch would look like shown in Fig. 15(a). With one shorting pin placed at the opposite side of the feed, the current density plot on the patch at a frequency different from the passband is shown in Fig. 15(b). One can conclude that RN can be developed at that frequency. In this way, additional shorting pins could be added to sharpen the filtering response when the out-of-band rejection achieved by a single RN is insufficient.

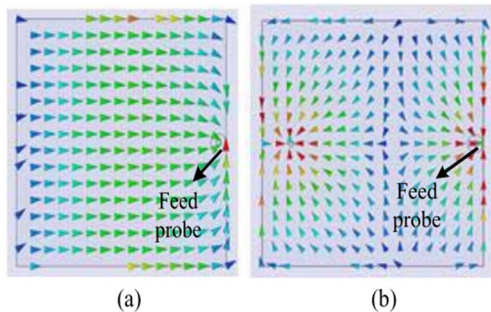


FIGURE 15. Surface current distribution with and without shorting pin in (a) and (b) respectively [4].

Now the SIW filtenna design used in [31] is considered. The design is shown in Fig. 16. The four metalized posts

utilized divide the cavity into two TE_{110} modes (usually TE_{120} in the absence of the slot) and also help in achieving inter-cavity coupling. The radiation slot again divides the right half cavity into half-mode resonators. Totally three resonant modes are formed and the design with just a single substrate acts like a third-order bandpass filter.

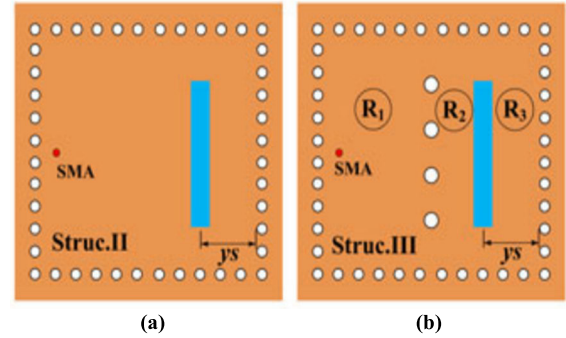


FIGURE 16. Slotted Cavity with and without metalized posts in (a) and (b) respectively [31].

At the resonant frequencies, the current near the slot is stronger resulting in radiation. At the RN frequency, at the location of the slot, the current is minimum/zero as shown in Fig. 17 resulting in the formation of RN. The position of the slot can be varied to control the RN. The S_{11} plot, realized Gain and antenna efficiency plots are shown in Fig. 18. One controllable RN, towards the lower edge, is seen to be achieved.

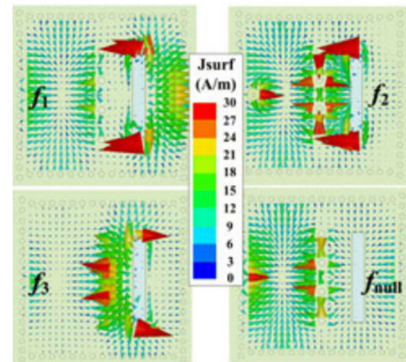


FIGURE 17. Plots of surface current around the slots at different frequencies [31].

F. USING MULTI-COUPPLING PATH BETWEEN SOURCE/LOAD AND RESONATOR - 1

The filtenna used in [43] makes use of multiple coupling paths between cavities and load to achieve RNs. A 3D view of the dual-layer structure is shown in Fig. 19.

MIM capacitive loading perturbations control TE_{210} and TE_{120} modes in the upper cavity. The lower cavity is slot-loaded. An inductive iris window helps in giving coupling between the cavities. The individual cavity coupling schemes are shown in Fig. 20. The design exhibits dual frequency

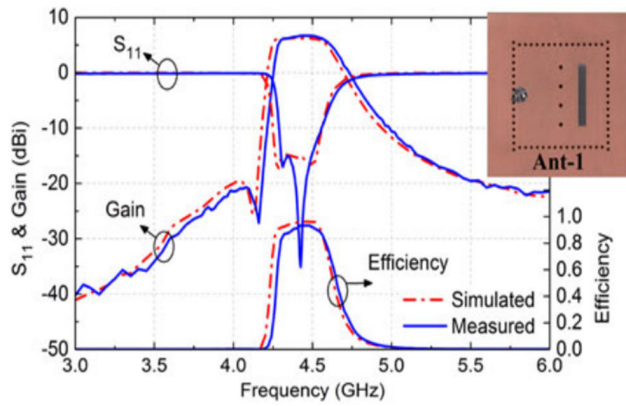


FIGURE 18. Plots of realized Gain and S_{11} and efficiency [31].

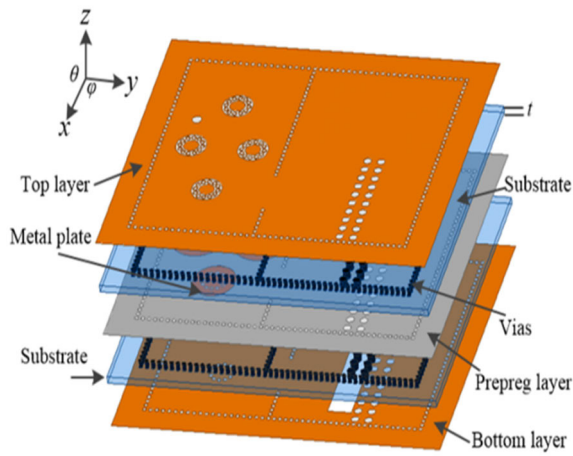


FIGURE 19. Exploded view of the filtenna with multiple coupling paths [43].

response and its coupling diagrams for two cases are shown in Fig. 21.

From [43, pp. 126], “the passband f_1 (f_2) is formed by the TE_{210} (TE_{120}) mode in the first cavity and TE_{110} (TE_{210}) mode in the second cavity”. These resonant cavities undergo coupling in such a way that RNs are formed. In Fig. 22, S_{11} and Gain plots of the design are shown. RNs are achieved at both band edges of the second resonant frequency. A good lower band edge/upper band edge rejection level of 30 dB/20 dB is seen.

G. USING MULTI-COUPLING PATH BETWEEN SOURCE/LOAD AND RESONATOR - 2

In [44], the filtenna (shown in Fig. 23) is made up of two cavities that are inter-coupled. The first cavity carries two modes TE_{201} and TE_{101} , designated (1) and (2) in the coupling diagram shown in Fig. 24. These modes are coupled to the second cavity where a slot divides it into two halves, setting up two half modes shown as (3) and (4), in the diagram.

An inductive iris window is present between the two cavities for coupling. Considering the two paths available

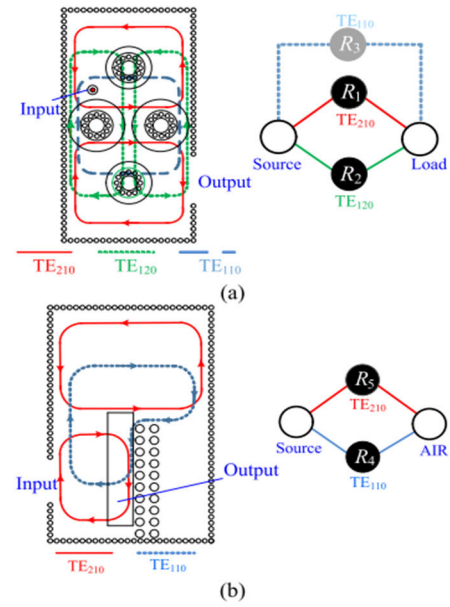


FIGURE 20. Magnetic field distributions and coupling in (a) the upper cavity and (b) the radiation cavity [43].

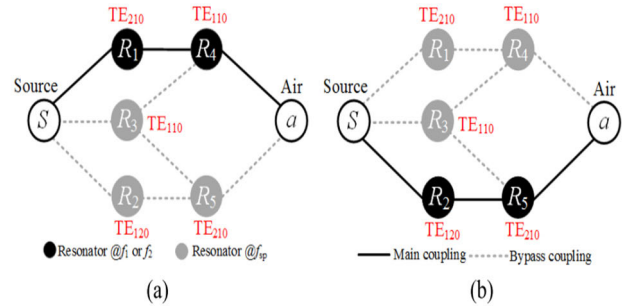


FIGURE 21. Coupling diagram of the filtenna at (a) low band and (b) high band [43].

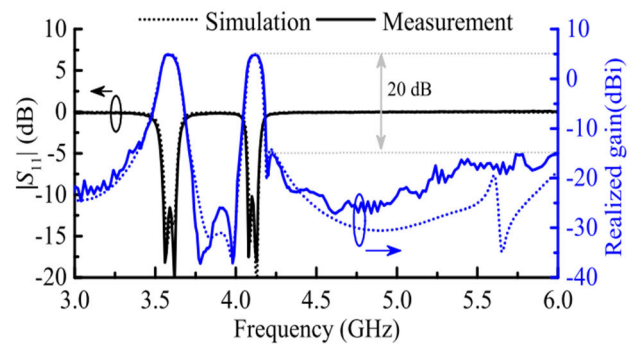


FIGURE 22. Measured and simulated realized Gain and S_{11} plots [43].

from source to resonator 3, S-1-3 and S-2-3 the coupling coefficients, M_{S2} and M_{23} are out-of-phase, which results in the generation of one RN on the higher side of the pass band. Changing the feed line position helps in controlling this RN.

Due to the cancellation of the signal along the paths (3-Load) and (3-4-Load), RN on the lower side is generated. The slot's position can be varied to control this RN. The

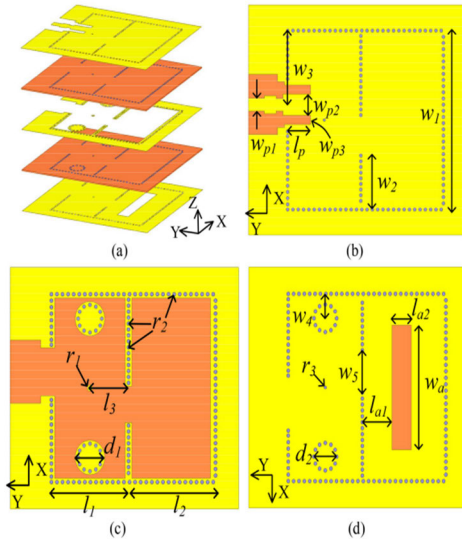


FIGURE 23. Structure of the filtenna employing multiple coupling paths (a) 3D view (b) top metal layer (c) middle layer (d) bottom layer [44].

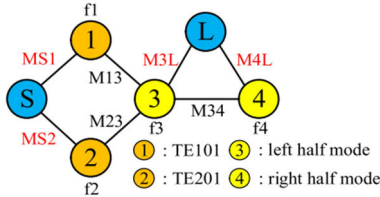


FIGURE 24. Coupling diagram of the filtenna [44].

two RNs can be seen in the Gain graph shown in Fig. 25. The rejection level towards the upper band edge is better compared to the lower edge.

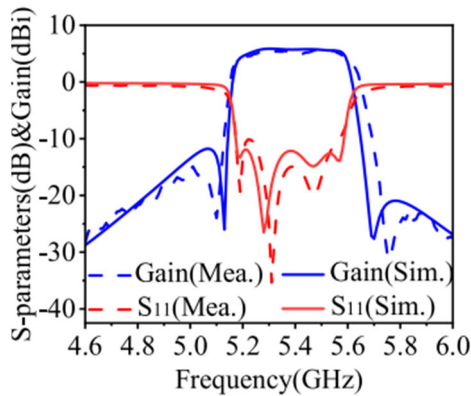


FIGURE 25. Measured results of the filtenna [44].

H. USING ELECTRIC AND MAGNETIC MIXED-COUPLING - 1

As noted by the authors in [48, p. 431], “for two coupled resonators, the coupled sides with the maximum magnetic field are combined with their open gaps with the maximum electric field, creating the electric and magnetic mixed-coupling”.

The coupling structure used in [46] is a stepped impedance coupling structure with loaded stubs. The design is shown in Fig. 26.

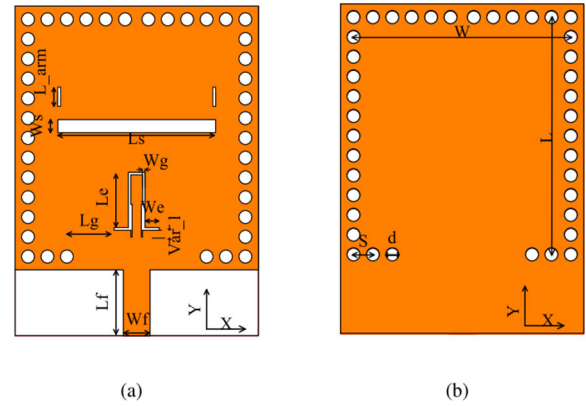


FIGURE 26. Top and bottom views, shown in (a) and (b) respectively, of the filtenna with slot and Electric and Magnetic mixed-coupling structure [46].

The structure for achieving mixed-coupling is located below the perturbation slot. It has a stepped impedance transmission line (vertically oriented) to achieve electric coupling and the parameter, L_e controls its strength. The horizontally oriented ones along with slot stubs having length Var_1 achieve magnetic coupling. The structure produces a RN because of the mixed coupling. Variation of the RN with the variable Var_1 is shown in Fig. 27.

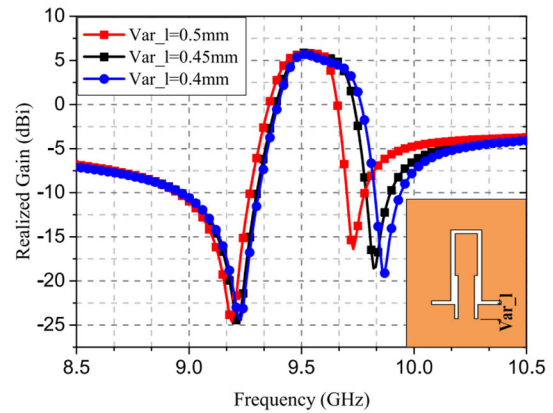


FIGURE 27. Variation of RN with Var_1 parameter [46].

One RN (towards the lower edge) is formed by the fundamental mode's resonance. RN at the other band edge is formed due to electric and magnetic mixed-coupling. The circuit equivalent of the mixed coupling is shown in Fig. 28. It presents a transmission zero or indirectly a RN.

I. USING ELECTRIC AND MAGNETIC MIXED-COUPLING - 2

The filtenna used in [47] is composed of a cavity resonator integrated with a radiating patch (see Fig. 29).

We know from the resonant theory of [71], that for the mixed electric and magnetic coupling, when

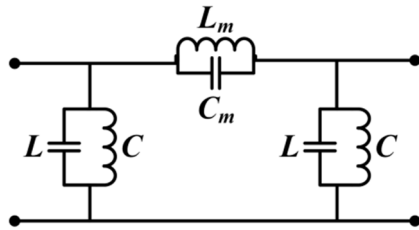


FIGURE 28. Equivalent of EM mixed coupling [46].

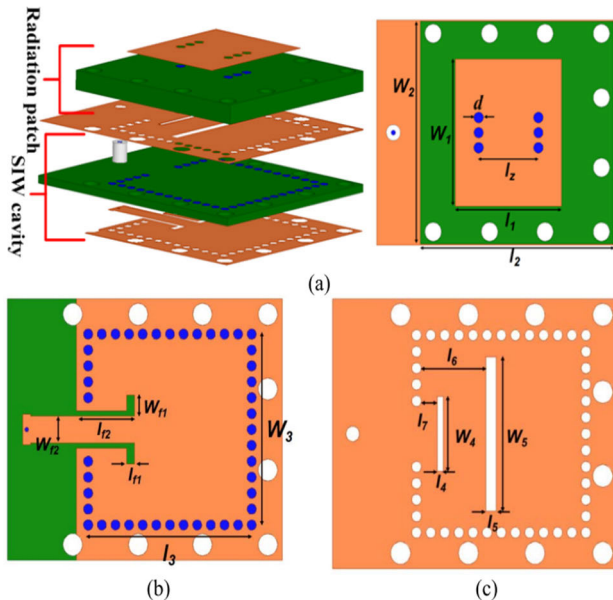


FIGURE 29. (a) 3D view and (b), (c) layered views of the filtenna [47].

$f < f_0$, it can be equivalent to an inductance (magnetic) coupling. When $f > f_0$, it can be equivalent to a capacitance (electric) coupling [47, p. 2337].

The cavity resonator has two asymmetric slots etched on it to couple with the radiating patch at the top. The longer slot divides the cavity into two half- TE_{110} modes. A shorter slot is positioned such that the existing magnetic coupling changes to a mixed electric and magnetic coupling. This would introduce a RN. S_{11} and the Gain plots are shown in Fig. 30. A 9 dB gain is seen in the passband with a significant level of suppression outside the band. The circuit equivalent of the filtenna is shown in Fig. 31.

J. USING PARASITIC STRIP

In [32] the metallic U-shaped strip placed in between the two substrates along with the probe feed (refer Fig. 32) forms resonance at a frequency close to the main resonant frequency which helps in broadening of bandwidth. A more useful outcome of this is the formation of a RN at the lower side of the passband, helping in increasing the frequency selectivity. Slots present on the patch give RN at the upper side.

At the RN frequency surface current concentrates only in the middle of the patch. Along the metal strip, the distributed currents cancel yielding a null.

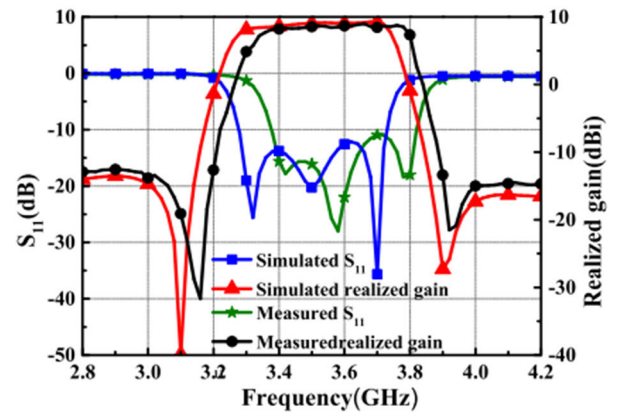


FIGURE 30. S_{11} and Gain results of the filtenna [47].

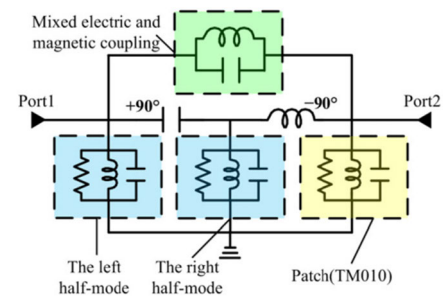


FIGURE 31. Circuit model of the filtenna [47].

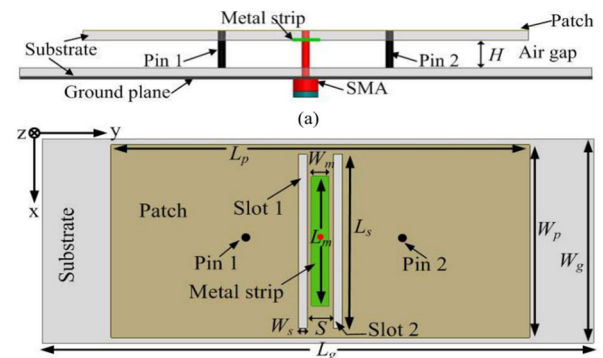


FIGURE 32. Filtenna using a metallic strip and slots to achieve RN [32].

Two radiation slots are introduced on the patch symmetrically to the feed at positions where TM_{02} mode currents are weaker. The resulting current plot is shown in Fig. 33. On both sides of the slot, we get out-of-phase currents which gives a null in the far field. The RNs for the design are seen at 2.8 GHz and 4 GHz on either side of resonance as shown in the Gain plot in Fig. 34.

K. USING A STACKED PATCH AND T-SHAPED STRIP

The filtenna described in [52] is a 2 by 2 filtenna array. Each element has a three-layer design presenting three resonances and four RNs. The elemental design is shown in Fig. 35. A U-shaped slot is etched on the patch present at the first layer. The second layer holds the main patch that has a folded

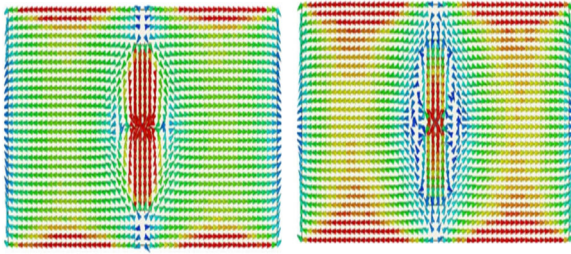


FIGURE 33. Surface current plots around the strip and slots [32].

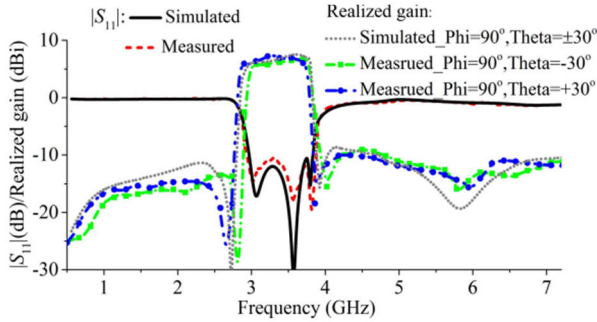


FIGURE 34. Plots of S_{11} and realized Gain for the design [32].

U-shaped slot. There exist parasitic patches on either side of this main patch. Refer to Fig. 36. And on the third layer is printed a folded T-shaped strip.

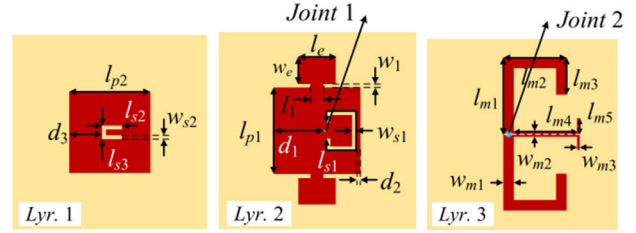


FIGURE 36. Construction detail of each layer [52].

U-shaped slot present on layer 2. The plot of surface current density on the layer confirms the same. The current flow is in opposite directions on either side of the slot.

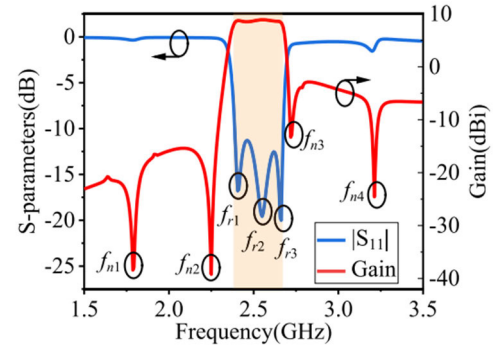


FIGURE 37. S_{11} and Gain graphs showing the four RNs [52].

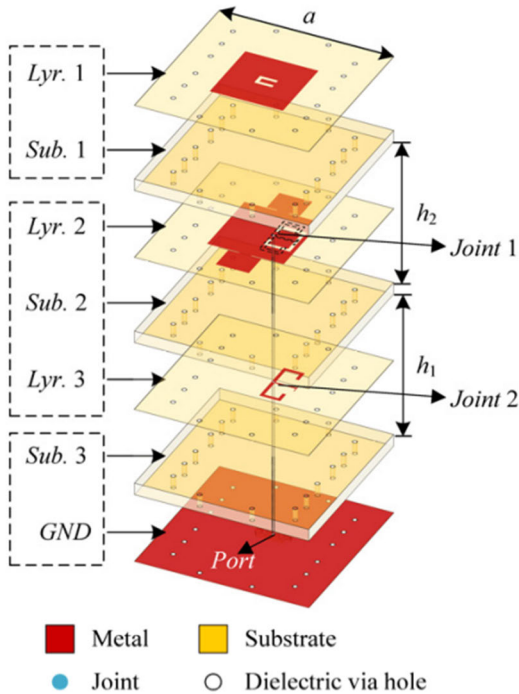


FIGURE 35. Configuration of the filtenna [52].

Fig. 37 shows the S parameter and Gain graphs. The two RNs towards the lower band edge are named f_{n1} and f_{n2} , and the ones towards the upper edge are named f_{n3} and f_{n4} . The radiation null, f_{n2} , observed from Fig. 38(b), results due to the

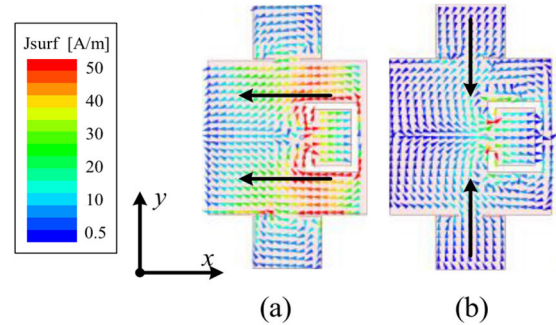


FIGURE 38. The plot of surface current on layer 2 at (a) a resonant frequency and (b) RN f_{n2} [52].

Fig. 39 shows the current distribution plots on the first and the second layers. Analyzing the current flow given in Figs. 39(b) and 39(c) the formation of the RN f_{n3} is understood.

The other two RNs are formed because of the folded T-shaped strip. Currents on the main patch and the strip being in opposite directions at both f_{n1} and f_{n4} result in two more RNs. The current distributions are shown in Fig. 40.

L. ACHIEVING RADIATION NULL AT FEED ITSELF

The filtenna described in [54] is shown in Fig. 41. It utilizes a Uniform Impedance Resonator (UIR) near the feeding circuit.

The circuit equivalent of the UIR is shown in Fig. 42(b). The annular strip lines present inductance L_1 and capacitance C . The arc-shaped vias present additional inductance L_2 .

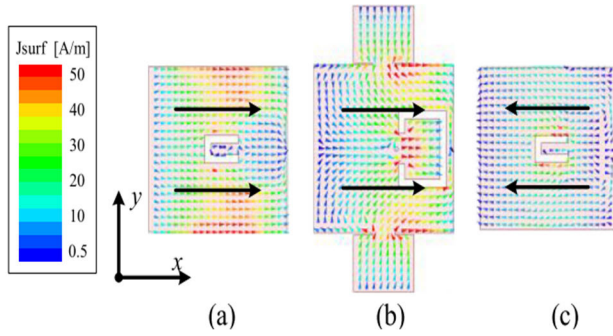


FIGURE 39. The plot of surface current on (a) layer 1 and (b) layer 2 at a resonant frequency and (c) layer 1 at RN f_{n3} [52].

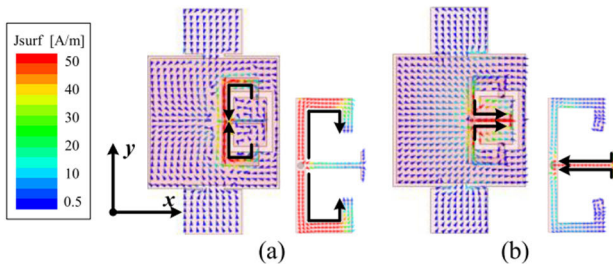


FIGURE 40. The plot of surface current on layer 2 and layer 3 at (a) RN f_{n1} and (b) RN f_{n4} [52].

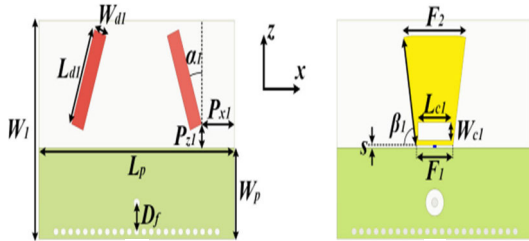


FIGURE 41. Top and bottom views of the filtenna from [54].

The UIR produces one resonance and a radiation null simultaneously as can be seen in Fig. 43.

An open-circuit stub positioned on the feeding structure generates a radiation null applicable to the whole antenna.

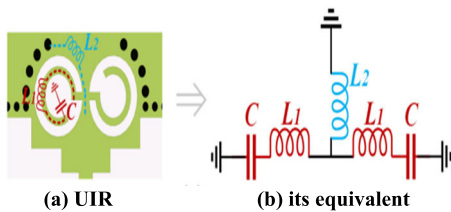


FIGURE 42. Expanded viewed the UIR and its equivalent [54].

III. COMPARISON

The designs that are reviewed are compared in Table 1 with regard to the parameters: number of cavities, number of resonant modes, resonant frequency, realized Gain, number of controllable radiation nulls, out-of-band rejection seen at lower and upper band edges, and antenna efficiency.

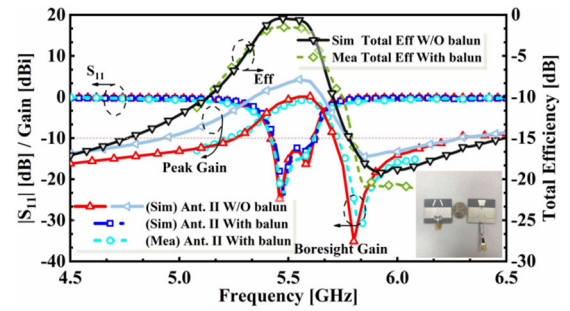


FIGURE 43. Plots of S_{11} , Gain, and efficiency of the filtenna from [54].

TABLE 1. Performance comparison of the designs discussed.

Ref	No. of cavities/ Filter order	Res. Freq (GHz)	Gain (dBi)	No. of CRNs	Out-of-band rejection (lower/upper edge) in dB	Eff. (%)
[28]	2/3	4.5	7.5	4	> 21.5/22.5	N/A
[29]	2/3	9.65	7.2	2	> 19/19	87
[30]	1/1	5.24	7.1	2	> 22/16	80
[34]	1/2	4.8, 5.85	7.88	2	> 19/20	95.6
[31]	1/3	4.4	6.7	1	> 26	94
[43]	2/4	3.6, 4.1	4.84	3	> 30/20	N/A
[44]	2/4	5.36	5.3	2	> 17.3/26	74
[46]	1/3	9.5	5.8	2	> 13/10	N/A
[47]	2/3	3.6	9.06	2	> 26/30	90
[32]	1/1	3.6	6.8	2	> 21/16	N/A
[52]	3/3	2.5	13	4	> 13.9/19.7	N/A
[54]	1/2	5.51	7.7	1	N/A	77

*Ref: Reference number, Eff: Antenna efficiency, CRN: Controllable Radiation Null

The designs in [28], [43], and [52] present more than two RNs. Stronger out-of-band rejection is seen in the designs, [31], [32], and [47]. Higher filter order is achieved in [43] and [44] using a lesser number of cavities. In [31], [34], and [47], larger antenna efficiency values are achieved.

IV. CONCLUSION

Some of the designs that provide improved frequency selectivity have been reviewed. Radiation nulls on either side of the passband can be achieved by using multiple coupling paths in filtennas, using slots of different shapes, shorting pins placed at appropriate positions, using Electric and Magnetic mixed-coupling structures, using stacked parasitic patches, loops, and using stepped impedance resonators. Having more than two RNs achieves better out-of-band rejection.

REFERENCES

- [1] Y. Lin Zhang, W. Hong, K. Wu, J. Xin Chen, and H. Jun Tang, "Novel substrate integrated waveguide cavity filter with defected ground structure," *IEEE Trans. Microwave Theory Techn.*, vol. 53, no. 4, pp. 1280–1287, Apr. 2005.

- [2] R. J. Cameron, "General coupling matrix synthesis methods for Chebyshev filtering functions," *IEEE Trans. Microwave Theory Techn.*, vol. 47, no. 4, pp. 433–442, Apr. 1999.
- [3] Q.-L. Zhang, W.-Y. Yin, S. He, and L.-S. Wu, "Compact substrate integrated waveguide (SIW) bandpass filter with complementary split-ring resonators (CSRRs)," *IEEE Microwave Wireless Compon. Lett.*, vol. 20, no. 8, pp. 426–428, Aug. 2010.
- [4] X. Y. Zhang, W. Duan, and Y.-M. Pan, "High-gain filtering patch antenna without extra circuit," *IEEE Trans. Antennas Propag.*, vol. 63, no. 12, pp. 5883–5888, Dec. 2015.
- [5] R. Lovato and X. Gong, "A third-order SIW-integrated filter/antenna using two resonant cavities," *IEEE Antennas Wireless Propag. Lett.*, vol. 17, pp. 505–508, 2018.
- [6] Y. Yusuf and X. Gong, "A new class of 3-D filter/antenna integration with high quality factor and high efficiency," in *Proc. IEEE MTT-S Int. Microwave Symp.*, 2010, pp. 892–895.
- [7] W. Wang, H. Jin, W. Yu, X. H. Zhang, F. Wu, K.-S. Chin, and G. Q. Luo, "A single-layer dual-circularly polarized SIW-cavity-backed patch filtenna with wide axial-ratio bandwidth," *IEEE Antennas Wireless Propag. Lett.*, vol. 20, pp. 908–912, 2021.
- [8] W. Menzel and A. Balalem, "Quasi-lumped suspended stripline filters and diplexers," *IEEE Trans. Microwave Theory Techn.*, vol. 53, no. 10, pp. 3230–3237, Oct. 2005.
- [9] H.-W. Xie, K. Zhou, C.-X. Zhou, and W. Wu, "Compact SIW diplexers and dual-band bandpass filter with wide-stopband performances," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 67, no. 12, pp. 2933–2937, Dec. 2020.
- [10] N. C. Pradhan, K. S. Subramanian, R. K. Barik, and Q. S. Cheng, "Design of compact substrate integrated waveguide based triple- and quad-band power dividers," *IEEE Microwave Wireless Compon. Lett.*, vol. 31, no. 4, pp. 365–368, Apr. 2021.
- [11] Y. Peng and L. Sun, "A compact broadband phase shifter based on HMSIW evanescent mode," *IEEE Microwave Wireless Compon. Lett.*, vol. 31, no. 7, pp. 857–860, Jul. 2021.
- [12] W. Hong, B. Liu, Y. Wang, Q. Lai, H. Tang, X. X. Yin, Y. D. Dong, Y. Zhang, and K. Wu, "Half mode substrate integrated waveguide: A new guided wave structure for microwave and millimeter wave application," in *Proc. 31st Int. Conf. Infr. Millim. Waves*, 2006, p. 219.
- [13] Z. Zhang, N. Yang, and K. Wu, "5-GHz bandpass filter demonstration using quarter-mode substrate integrated waveguide cavity for wireless systems," in *Proc. IEEE Radio Wireless Symp.*, 2009, pp. 95–98.
- [14] N. Grigoropoulos, B. Sanz-Izquierdo, and P. R. Young, "Substrate integrated folded waveguides (SIFW) and filters," *IEEE Microwave Wireless Compon. Lett.*, vol. 15, no. 12, pp. 829–831, Dec. 2005.
- [15] S. Moscato, N. Delmonte, L. Silvestri, M. Bozzi, and L. Perregrini, "Half-mode versus folded SIW filters modeling and design," in *Proc. IEEE MTT-S Int. Conf. Numer. Electromagn. Multiphys. Model. Optim.*, Aug. 2015, pp. 1–3.
- [16] G. H. Zhai, W. Hong, K. Wu, J. X. Chen, P. Chen, J. Wei, and H. J. Tang, "Folded half mode substrate integrated waveguide 3 dB coupler," *IEEE Microwave Wireless Compon. Lett.*, vol. 18, no. 8, pp. 512–514, Aug. 2008.
- [17] Y. Dong and T. Itoh, "Miniaturized substrate integrated waveguide slot antennas based on negative order resonance," *IEEE Trans. Antennas Propag.*, vol. 58, no. 12, pp. 3856–3864, Dec. 2010.
- [18] A. M. Hamzah, L. Audah, N. Alkhafaji, H. J. M. Albattat, and A. A. Abdulbari, "Substrate integrated waveguide SIW technology-based miniaturization and performance enhancement of antennas: A review," *Int. J. Adv. Sci. Technol.*, vol. 29, no. 6, pp. 1739–1754, 2020.
- [19] Y. D. Dong, T. Yang, and T. Itoh, "Substrate integrated waveguide loaded by complementary split-ring resonators and its applications to miniaturized waveguide filters," *IEEE Trans. Microwave Theory Techn.*, vol. 57, no. 9, pp. 2211–2223, Sep. 2009.
- [20] J. D. Martinez, S. Sirci, M. Taroncher, and V. E. Boria, "Compact CPW-fed combline filter in substrate integrated waveguide technology," *IEEE Microwave Wireless Compon. Lett.*, vol. 22, no. 1, pp. 7–9, Jan. 2012.
- [21] S. Ji, Y. Dong, and Y. Fan, "Compact filtering antennas based on dual-mode SIW cavity," in *IEEE MTT-S Int. Microwave Symp. Dig.*, Nov. 2021, pp. 50–52.
- [22] J.-Y. Yin, T.-L. Bai, J.-Y. Deng, J. Ren, D. Sun, Y. Zhang, and L.-X. Guo, "Wideband single-layer substrate integrated waveguide filtering antenna with U-Shaped slots," *IEEE Antennas Wireless Propag. Lett.*, vol. 20, pp. 1726–1730, 2021.
- [23] M.-H. Ho, J.-C. Li, and Y.-C. Chen, "Miniaturized SIW cavity resonator and its application in filter design," *IEEE Microwave Wireless Compon. Lett.*, vol. 28, no. 8, pp. 651–653, Aug. 2018.
- [24] R. Ruf and W. Menzel, "A novel compact suspended stripline resonator," *IEEE Microwave Wireless Compon. Lett.*, vol. 22, no. 9, pp. 444–446, Sep. 2012.
- [25] H.-T. Hu, K. F. Chan, and C. H. Chan, "60 GHz Fabry-Pérot cavity filtering antenna driven by an SIW-fed filtering source," *IEEE Trans. Antennas Propag.*, vol. 70, no. 2, pp. 823–834, Feb. 2022.
- [26] K.-Z. Hu, H.-Y. Huang, M.-C. Tang, Z. Chen, D. Yan, and P. Wang, "A single-layer wideband differential-fed filtering antenna with high selectivity," *IEEE Trans. Antennas Propag.*, vol. 71, no. 5, pp. 4522–4527, May 2023.
- [27] D. Pang and L. Li, "A low-profile and low H-plane cross polarization filter shorted patch antenna with high selectivity," *Int. Appl. Comput. Electromagn. Soc.*, vol. 1, pp. 1–2, Aug. 2019.
- [28] Y. Zhang, X. Y. Zhang, and Q. H. Liu, "A dual-layer filtering SIW slot antenna utilizing double slot coupling scheme," *IEEE Antennas Wireless Propag. Lett.*, vol. 20, pp. 1073–1077, 2021.
- [29] X. Liu, X. Zhang, K. Xu, and J. Shi, "A filtering antenna with high frequency selectivity using stacked dual-slotted substrate integrated cavities," *IEEE Antennas Wireless Propag. Lett.*, vol. 19, pp. 1311–1315, 2020.
- [30] J. Y. Jin, S. Liao, and Q. Xue, "Design of filtering-radiating patch antennas with tunable radiation nulls for high selectivity," *IEEE Trans. Antennas Propag.*, vol. 66, no. 4, pp. 2125–2130, Apr. 2018.
- [31] P. K. Li, C. J. You, H. F. Yu, X. Li, Y. W. Yang, and J. H. Deng, "Codesigned high-efficiency single-layered substrate integrated waveguide filtering antenna with a controllable radiation null," *IEEE Antennas Wireless Propag. Lett.*, vol. 17, pp. 295–298, 2018.
- [32] J.-F. Li, Z. N. Chen, D.-L. Wu, G. Zhang, and Y.-J. Wu, "Dual-beam filtering patch antennas for wireless communication application," *IEEE Trans. Antennas Propag.*, vol. 66, no. 7, pp. 3730–3734, Jul. 2018.
- [33] D. Li and C. Deng, "A single-layer filtering antenna with two controllable radiation nulls based on the multimodes of patch and SIW resonators," *IEEE Antennas Wireless Propag. Lett.*, vol. 22, pp. 551–555, 2023.
- [34] W. Yang, Y. Zhang, W. Che, M. Xun, Q. Xue, G. Shen, and W. Feng, "A simple, compact filtering patch antenna based on mode analysis with wide out-of-band suppression," *IEEE Trans. Antennas Propag.*, vol. 67, no. 10, pp. 6244–6253, Oct. 2019.
- [35] L. Li, S. Wu, D. Pang, X. Zhang, and Q. Wang, "A fifth-order single-layer dual-band half-mode SIW filtering antenna with a multifunctional single slot," *IEEE Antennas Wireless Propag. Lett.*, vol. 20, pp. 1676–1680, 2021.
- [36] K.-Z. Hu, M.-C. Tang, D. Li, Y. Wang, and M. Li, "Design of compact, single-layered substrate integrated waveguide filtenna with parasitic patch," *IEEE Trans. Antennas Propag.*, vol. 68, no. 2, pp. 1134–1139, Feb. 2020.
- [37] Q. Liu, L. Zhu, J. Wang, and W. Wu, "A wideband patch and SIW cavity hybrid antenna with filtering response," *IEEE Antennas Wireless Propag. Lett.*, vol. 19, pp. 836–840, 2020.
- [38] Z. Zheng, D. Li, X. Tan, and Q. Chen, "Single-layer dual-/tri-band SIW filtenna based on multifunctional cavity-backed slots," *IEEE Trans. Antennas Propag.*, vol. 71, no. 5, pp. 4498–4503, May 2023.
- [39] Y. Zhang, W. Yang, W. Che, Q. Xue, and L. Chen, "A wideband filtering patch antenna with multiple radiation nulls for good stopband suppression," in *Proc. IEEE MTT-S Int. Wireless Symp.*, May 2019, pp. 1–3.
- [40] K.-Z. Hu, M.-C. Tang, Y. Wang, D. Li, and M. Li, "Compact, vertically integrated duplex filtenna with common feeding and radiating SIW cavities," *IEEE Trans. Antennas Propag.*, vol. 69, no. 1, pp. 502–507, Jan. 2021.
- [41] B. Zhang and Q. Xue, "Filtering antenna with high selectivity using multiple coupling paths from source/load to resonators," *IEEE Trans. Antennas Propag.*, vol. 66, no. 8, pp. 4320–4325, Aug. 2018.
- [42] K.-R. Xiang, F.-C. Chen, Q. Tan, and Q.-X. Chu, "High-selectivity filtering patch antennas based on multipath coupling structures," *IEEE Trans. Microwave Theory Techn.*, vol. 69, no. 4, pp. 2201–2210, Apr. 2021.
- [43] D. Zhao, F. Lin, H. Sun, and X. Y. Zhang, "A miniaturized dual-band SIW filtering antenna with improved out-of-band suppression," *IEEE Trans. Antennas Propag.*, vol. 70, no. 1, pp. 126–134, Jan. 2022.
- [44] H.-Y. Xie, B. Wu, Y.-L. Wang, C. Fan, J.-Z. Chen, and T. Su, "Wideband SIW filtering antenna with controllable radiation nulls using dual-mode cavities," *IEEE Antennas Wireless Propag. Lett.*, vol. 20, pp. 1799–1803, 2021.

- [45] S. Fu, J. Su, and X. Chen, "Design of high selectivity filtering antenna based on multi-coupling path," in *Proc. Cross Strait Quad-Regional Radio Sci. Wireless Technol. Conf. (CSQRWC)*, Jul. 2019, pp. 1–2.
- [46] C. Wang, X. Wang, H. Liu, Z. Chen, and Z. Han, "Substrate integrated waveguide filtenna with two controllable radiation nulls," *IEEE Access*, vol. 8, pp. 120019–120024, 2020.
- [47] C. Fan, B. Wu, Y.-L. Wang, H.-Y. Xie, and T. Su, "High-gain SIW filtering antenna with low H-plane cross polarization and controllable radiation nulls," *IEEE Trans. Antennas Propag.*, vol. 69, no. 4, pp. 2336–2340, Apr. 2021.
- [48] Q.-X. Chu and H. Wang, "A compact open-loop filter with mixed electric and magnetic coupling," *IEEE Trans. Microwave Theory Techn.*, vol. 56, no. 2, pp. 431–439, Feb. 2008.
- [49] L. Wang, K. Xu, and J. Shi, "The methods for generating radiation null in filtering antennas," in *Proc. Photon. Electromagn. Res. Symp. Fall*, Dec. 2019, pp. 1531–1536.
- [50] J. Wu, Z. Zhao, Z. Nie, and Q.-H. Liu, "A printed unidirectional antenna with improved upper band-edge selectivity using a parasitic loop," *IEEE Trans. Antennas Propag.*, vol. 63, no. 4, pp. 1832–1837, Apr. 2015.
- [51] N. Nie and Z.-H. Tu, "Wideband filtering dumbbell-shaped slot antenna with improved frequency selectivity for both band-edges," *IEEE Access*, vol. 8, pp. 121479–121485, 2020.
- [52] F. Wei, X. Liu, X.-Z. Ding, X.-B. Zhao, and P.-Y. Qin, "A balanced filtering antenna array with high gain, steep selectivity, and multiradiation nulls parallel-fed by differential broadband network," *IEEE Trans. Antennas Propag.*, vol. 71, no. 12, pp. 9926–9931, Dec. 2023.
- [53] F. Wei, X.-B. Zhao, and X. W. Shi, "A balanced filtering quasi-yagi antenna with low cross-polarization levels and high common-mode suppression," *IEEE Access*, vol. 7, pp. 100113–100119, 2019.
- [54] Y. Xu, Y. Zhu, S. Wen, and Y. Dong, "Planar quasi-isotropic antenna and its implementation of filtering response," *IEEE Antennas Wireless Propag. Lett.*, vol. 20, pp. 2407–2411, 2021.
- [55] M. Li, S. Tian, M.-C. Tang, and L. Zhu, "A compact low-profile hybrid-mode patch antenna with intrinsically combined self-decoupling and filtering properties," *IEEE Trans. Antennas Propag.*, vol. 70, no. 2, pp. 1511–1516, Feb. 2022.
- [56] H.-T. Hu and C. H. Chan, "Substrate-integrated-waveguide-fed wideband filtering antenna for MillimeterWave application," *IEEE Trans. Antennas Propag.*, vol. 69, no. 12, pp. 8125–8135, Dec. 2021.
- [57] Q. Wu, J. Yin, C. Yu, H. Wang, and W. Hong, "Broadband planar SIW cavity-backed slot antennas aided by unbalanced shorting vias," *IEEE Antennas Wireless Propag. Lett.*, vol. 18, pp. 363–367, 2019.
- [58] K.-Z. Hu, M.-C. Tang, M. Li, and R. W. Ziolkowski, "Compact, low-profile, bandwidth-enhanced substrate integrated waveguide filtenna," *IEEE Antennas Wireless Propag. Lett.*, vol. 17, pp. 1552–1556, 2018.
- [59] X. Zhang, T.-Y. Tan, Q.-S. Wu, L. Zhu, S. Zhong, and T. Yuan, "Pin-loaded patch antenna fed with a dual-mode SIW resonator for bandwidth enhancement and stable high gain," *IEEE Antennas Wireless Propag. Lett.*, vol. 20, pp. 279–283, 2021.
- [60] A. Iqbal, I. B. Mabrouk, M. Al-Hasan, M. Nedil, and T. A. Denidni, "Wideband substrate integrated waveguide antenna for full-duplex systems," *IEEE Antennas Wireless Propag. Lett.*, vol. 21, pp. 212–216, 2022.
- [61] G. Qing Luo, Z. Fang Hu, Y. Liang, L. Yang Yu, and L. Ling Sun, "Development of low profile cavity backed crossed slot antennas for planar integration," *IEEE Trans. Antennas Propag.*, vol. 57, no. 10, pp. 2972–2979, Oct. 2009.
- [62] K. Yoshihara, M. Tamura, Y. Miyaji, and A. Alphones, "Dual band SIW cavity-backed crossed-slot antenna," in *Proc. IEEE Asia-Pacific Microwave Conf.*, vol. 19, May 2019, pp. 1539–1541.
- [63] K. Dhawaj, H. Tian, and T. Itoh, "Low-profile dual-band filtering antenna using common planar cavity," *IEEE Antennas Wireless Propag. Lett.*, vol. 17, pp. 1081–1084, 2018.
- [64] S. Ji, Y. Dong, and Y. Fan, "Low-profile dual-band filtering antenna with a shared SIW cavity," *IEEE Antennas Wireless Propag. Lett.*, vol. 20, pp. 2053–2057, 2021.
- [65] R. Lu, C. Yu, Y. Zhu, X. Xia, and W. Hong, "Millimeter-wave dual-band dual-polarized SIW cavity-fed filtenna for 5G applications," *IEEE Trans. Antennas Propag.*, vol. 70, no. 11, pp. 10104–10112, Nov. 2022.
- [66] A. K. Nayak, M. Chouhan, K. Moez, M. V. Kartikeyan, and A. Patnaik, "Dual-band substrate integrated waveguide (SIW) cavity slot antenna for C-band applications," in *Proc. IEEE Microwave, Antennas, Propag. Conf. (MAPCON)*, Dec. 2022, pp. 1302–1307.
- [67] M. Bozzi, L. Perregrini, and K. Wu, "Modeling of conductor, dielectric, and radiation losses in substrate integrated waveguide by the boundary integral-resonant mode expansion method," *IEEE Trans. Microwave Theory Techn.*, vol. 56, no. 12, pp. 3153–3161, Dec. 2008.
- [68] E. Massoni, M. Bozzi, and K. Wu, "Increasing efficiency of leaky-wave antenna by using substrate integrated slab waveguide," *IEEE Antennas Wireless Propag. Lett.*, vol. 18, pp. 1596–1600, 2019.
- [69] Y. Yusuf, H. Cheng, and X. Gong, "A seamless integration of 3-D vertical filters with highly efficient slot antennas," *IEEE Trans. Antennas Propag.*, vol. 59, no. 11, pp. 4016–4022, Nov. 2011.
- [70] A. A. Diman, F. Karami, P. Rezaei, A. Amn-E-Elahi, Z. Mousavirazi, T. A. Denidni, and A. A. Kishk, "Efficient SIW-feed network suppressing mutual coupling of slot antenna array," *IEEE Trans. Antennas Propag.*, vol. 69, no. 9, pp. 6058–6063, Sep. 2021.
- [71] R. J. Cameron, C. M. Kudsia, and R. R. Mansour, *Microwave Filters for Communication Systems*. Hoboken, NJ, USA: Wiley, 2018.



A. VEERABHADRA RAO (Graduate Student Member, IEEE) received the B.Tech. degree in electronics and communication engineering from the College of Engineering Kakinada, Jawaharlal Nehru Technological University Hyderabad, and the M.E. degree in systems and signal processing from the University College of Engineering, Osmania University, Hyderabad, in 2011. He has been an Assistant Professor with the Department of Electronics and Communication Engineering, Mahatma Gandhi Institute of Technology, Gandipet, Hyderabad, since 2006. He is currently a Research Scholar with the Department of ECE, National Institute of Technology Warangal, India. His research interest includes compact filter-antenna designs with improved frequency selectivity.



GOPI RAM (Senior Member, IEEE) received the B.E. degree in electronics and telecommunication engineering from the Government Engineering College Jagdalpur, Jagdalpur, Chhattisgarh, India, in 2007, and the M.Tech. degree in telecommunication engineering from the National Institute of Technology Durgapur, Durgapur, West Bengal, India, in 2011. In 2012, he joined as a full-time Institute Research Scholar with the National Institute of Technology Durgapur, to carry out research for the Ph.D. degree. He joined NIT Warangal, in April 2018, as an Assistant Professor with the Department of ECE, where he is currently an Assistant Professor. He has published more than 90 research papers in international journals and conferences. His research interests include the analysis and synthesis of antenna arrays via bio-inspired evolutionary algorithms and antenna array optimization of various radiation characteristics. He received the Scholarship from the Ministry of Human Resource and Development (MHRD), Government of India, from 2009 to 2011 (M.Tech. degree) and from 2012 to 2016 (Ph.D. degree).

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